MathsJam Shout

June 2026

Glasgow MathsJam, UK

Puzzle Clan Decline

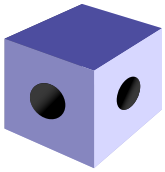
It is 1746. A single MacAshing chief survives in Scotland, and with him the clan name. Each generation, every man carrying the name independently fathers sons who inherit it; if any generation has none, the name is gone forever.

Each man fathers sons one at a time. Before each son, including the first, he fathers one with probability ⅔ and stops forever with probability ⅓ (so he has no sons at all with probability ⅓, exactly one son with probability ⅔ × ⅓ etc). All families are independent.

What is the probability the MacAshing name eventually dies out?

Hint: Express the probability of dying out, P(dying out), in terms of an expression also involving P(dying out). Can this expression be written in a closed form way?

Make Holey Cube



Imagine a cube is made of infinitely stretchable clay. You may bend, squash and morph it however you like, as long as you never cut it or glue parts together. A 'hole' means something you could thread a loop of string through so that it can't slip off.

Start with a solid cube and drill straight square tunnels clean through it.
Cube A: Two tunnels, left-right and front-back, at different heights so they miss each other.
Cube B: The same two tunnels at the same height, so they cross in the centre.
Cube C: Three tunnels, left-right, front-back, top-bottom, all through the centre, crossing in a single shared region.
Cube D: The same three directions, but offset so that no two tunnels ever meet.

How many holes does each cube have?
Hint: Two of the cubes have the same number of holes despite having a different number of tunnels.

Bring some plasticine to the pub and try to make the cubes yourself, using a straw or pencil to punch out holes!

Think Sun Fun

Model the Earth as a perfect sphere with its axis tilted 23.44° from the vertical to its orbit. Treat the Sun as so far away that its rays arrive parallel.

You are at the exact moment of the June solstice when the Earth's axis leans most toward the Sun.

1. At local solar noon, how high above the horizon is the Sun at your MathsJam?

2. Ancient people aligned monuments to the solstice sunrise. If you wanted to lay a stone avenue pointing exactly at where the midsummer Sun rises, which cardinal direction would it point?

3. Is there an aeroplane that you could take that would be fast enough to watch two sunsets, one first at Edinburgh MathsJam, then one at Glasgow MathsJam?

Prove Colouring In

The real plane is coloured with two different colours. Prove that, for any $D > 0$, there exist two points of the same colour at distance D apart.

More difficult: prove this for three colours.

(The four colour case was proved in 2018).

MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com**

Maths Jam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly Maths Jam nights happen in over 70 locations around the world, on the second-to-last Tuesday of each month. To find your nearest Maths Jam, visit the website at **mathsjam.com.**